

ARPIT GUPTA

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PROFESSIONAL SUMMARY

Final-year **Computer Science** undergraduate and full-stack developer with AI-integrated web applications built on **React, Next.js, Node.js, and Express**. Skilled in **REST API design, authentication (JWT, OAuth)**, and database architecture across **MongoDB, PostgreSQL, and Prisma**, with strong DSA fundamentals.

TECHNICAL SKILLS

Languages : JavaScript, TypeScript, Java, HTML, CSS, SQL

Frontend : React, Next.js, Tailwind CSS, Shadcn UI, React Router, Framer Motion, Zustand, CodeMirror

Backend : Node.js, Express.js, REST APIs, Bcrypt, JWT, HTTP Cookies, Authentication

Databases & ORM : MongoDB, Mongoose, PostgreSQL, Prisma

AI/LLM Integration : Google Gemini API, LLM Integration, Server-Sent Events (SSE), Prompt Engineering

Testing & Tools : Vitest, Postman, Git, GitHub, AWS, Vercel, Render, Clerk

PROJECTS

TutorialSync (AI Tutorial Modernizer) | [Live](#) | [GitHub](#)

React, Node.js, Express, MongoDB, Gemini AI

- Architected **TutorialSync** to fix stale tutorial syntax (e.g., Firebase v8 to v9, Next.js Pages Router to App Router), scanning videos and tech topics to **flag deprecated APIs** and generate modernized guides with side-by-side code comparisons.
- Hardened a **20-endpoint Node.js/Express REST API** integrating Google Gemini server-side, layering **JWT httpOnly cookies, Google OAuth, Helmet headers**, and **XSS/NoSQL-injection defense** (DOMPurify, Mongoose validation).
- Shipped a **React + Vite frontend** to production on Vercel and Render, featuring real-time analysis progress, syntax-highlighted "Then vs. Now" code comparisons, a persistent **AI Mentor chat** with saved history, and one-click Markdown export.

BigO.ai (Algorithmic Complexity Engine & Debugger) | [Live](#) | [GitHub](#)

React, Node.js, Express, Gemini AI

- Engineered a **2-tier complexity engine** pairing a **sub-1ms client-side static AST scanner** with a Gemini-powered deep verifier, delivering instant Big-O proofs, line-by-line debugger traces, and cross-language translation (JS, Python, C++, Java, Rust).
- Cut perceived wait time by **75%** by streaming Gemini responses via **Server-Sent Events**, backed by a SHA-256-keyed **in-memory LRU cache** resolving repeat analyses in **8ms** and an auto-repair parser for malformed LLM JSON output.
- Secured the platform with **bcrypt (cost 12)** hashing, CSPRNG-based OTPs, and **4-tier rate limiting**, validated by a **35-test Vitest suite**, and kept the free-tier backend warm 24/7 via a zero-cost auto-wake loop.

Lumina-HQ (Multi-Tenant Document Analysis Tool) | [Live](#) | [GitHub](#)

Next.js, Prisma, Gemini API, Clerk

- Designed **Lumina-HQ** from the ground up, a multi-tenant AI document intelligence platform on Next.js, Prisma, and PostgreSQL, exposing **5 REST API routes (6 endpoints)** across **5 Gemini-powered workflows**: summarization, Q&A, sentiment analysis, entity extraction, and structured JSON export.
- Eliminated cross-tenant data exposure risk with **workspace isolation** via Clerk Organizations and database-level query partitioning, serving files through **signed, time-limited Vercel Blob URLs** to block unauthorized downloads.
- Automated repetitive document review work with a **Gemini-powered processing pipeline** that turns unstructured documents into cited Q&A responses, executive summaries, extracted entities, and structured JSON outputs.

CERTIFICATIONS

- Oracle Certified Java Foundations Associate** — Certified in Java syntax, OOP principles, data types, and control flow.
- Oracle Agentic AI Certified Foundations Associate** — Validated in agentic AI architectures, autonomous decision-making, and workflow orchestration.
- IBM Cyber Security Analyst** — Trained in network security fundamentals, threat detection, and incident response.
- Google AI Essentials** — Completed hands-on training in generative AI tools and responsible AI practices.
- Google Prompting Essentials** — Practiced structured prompt design: context-setting, few-shot examples, and iterative refinement.

EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering

July 2023 – July 2027

CGPA: 8.72/10